DATA110 CAPSTONE ·

Engagement Efficiency Segmentation & Growth Prediction for Twitch Channels

Beyond raw follower counts: what engagement behavior actually predicts channel growth

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Raw metrics conflate size with engagement

A channel with 10M followers who watch a few minutes a month isn't the same as a channel with 100K followers watching for hours every week. Raw follower and watch-time figures don't distinguish between them.

Most public analyses of this exact dataset predict one raw metric from another (e.g. Followers from Watch Time) — which mostly re-learns that bigger channels have bigger numbers.

978 / 22

Partnered vs. Non-Partnered channels in this dataset

A classifier hits 97.8% accuracy just by always guessing "True" — without learning anything.

This is the kind of surface-level trap raw-metric analysis falls into — and why this project engineers deeper features instead.

OBJECTIVES

Two research questions

01 Can channels be segmented by engagement quality — independent of size — using engineered ratio features?

02 Do these engagement features predict absolute follower growth, and which behaviors matter most?

Answered via the advanced-level primary research project, built on a basic and intermediate baseline (per project structure).

DATASET

Top 1,000 Twitch channels, 11 fields

1,000	11	0	485
channels	raw fields	missing values	English channels

Watch time · Stream time · Peak / Average viewers · Followers · Followers gained · Views gained · Partnered · Mature · Language

Key data-quality decision

Partnered is severely imbalanced (978/22) — rejected as a classification target. Mature (770/230) is well-balanced and used instead.

A five-stage pipeline

1. Data Cleaning
2. Feature Engineering
3. Basic Level
4. Intermediate Level
5. Advanced Level

The advanced level is the primary research contribution — the other four stages build toward it.

Six engineered engagement ratios

Avg-to-Peak Ratio

Viewership consistency vs. one-off spikes

Watch Time / Follower

Cumulative attention per follower

Views / Follower

Reach efficiency vs. existing audience

Follower Growth Rate

Growth normalized by channel size

Viewers / Stream Hour

Audience density per hour streamed

Stickiness

Watch time beyond the expected baseline

Engagement Efficiency Score (EES)

Outliers first — a handful of channels had ratio values 5–40× larger than the rest, which would dominate scaling.

6 raw ratios → Winsorize (1st–99th pct.) → Min-Max scale [0,1] → Equal-weight average → **EES**

Why equal weighting? No prior theoretical basis favors one engagement dimension over another — a defensible, documented default.

Basic & Intermediate levels

BASIC · Exploratory Regression

Watch Time → Average Viewers

$R^2 = 0.406$

Watch time explains a moderate but incomplete share of viewership variance — motivating richer features later.

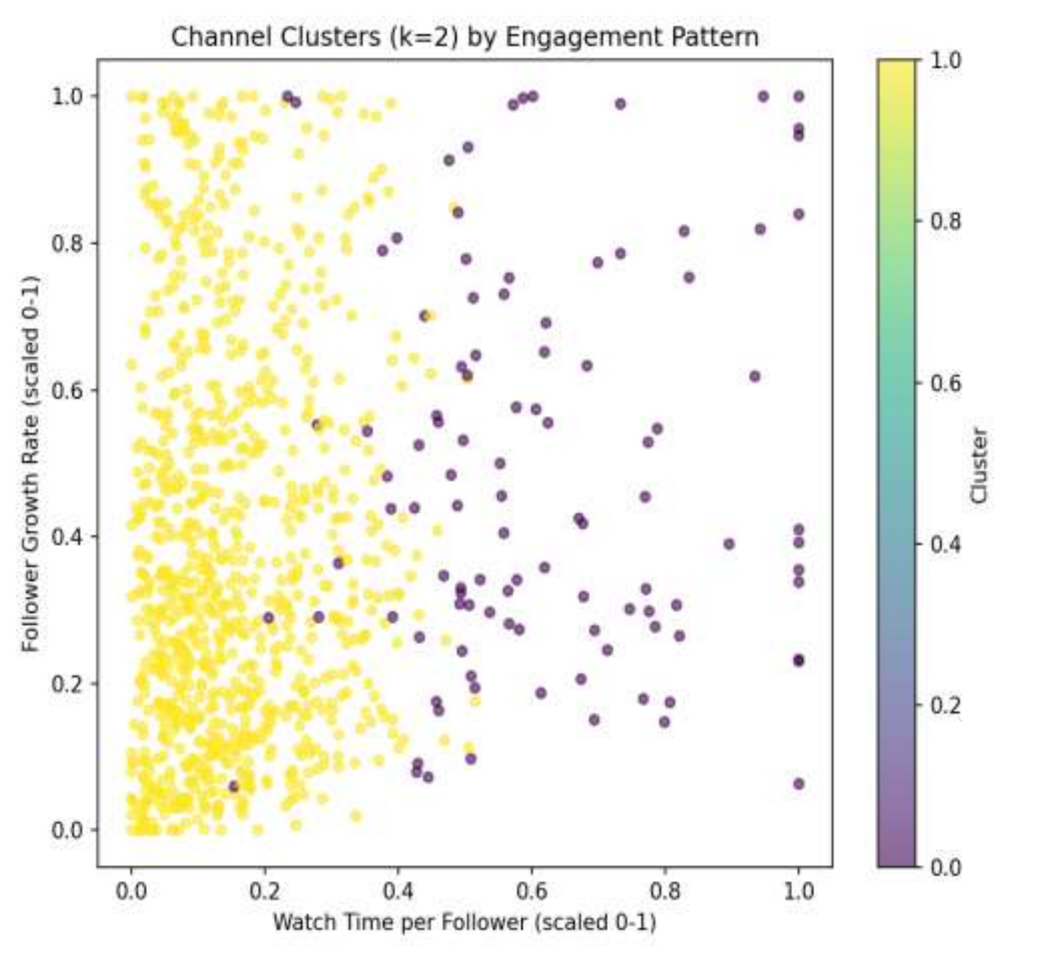
INTERMEDIATE · Classification

Predicting the Mature flag

Acc 0.475 · F1 0.348

Modest, and honestly so: raw engagement stats carry limited signal about content-type flags — a genuine finding, not a failure.

Segmenting channels by engagement



k = 2 selected by silhouette score

0.386 vs. 0.208–0.299 for k = 3–6

Cluster 0 — High-Efficiency

105 channels · mean EES = 0.338

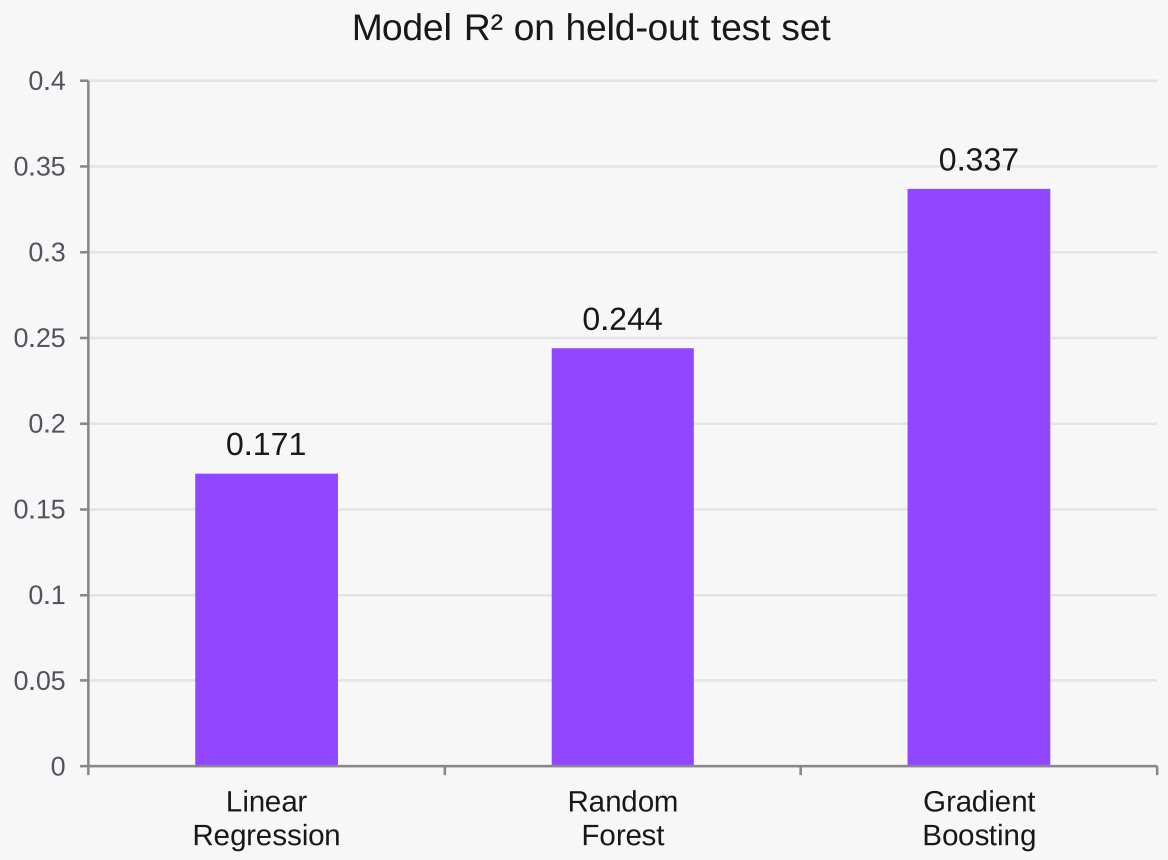
Niche, highly retained audiences

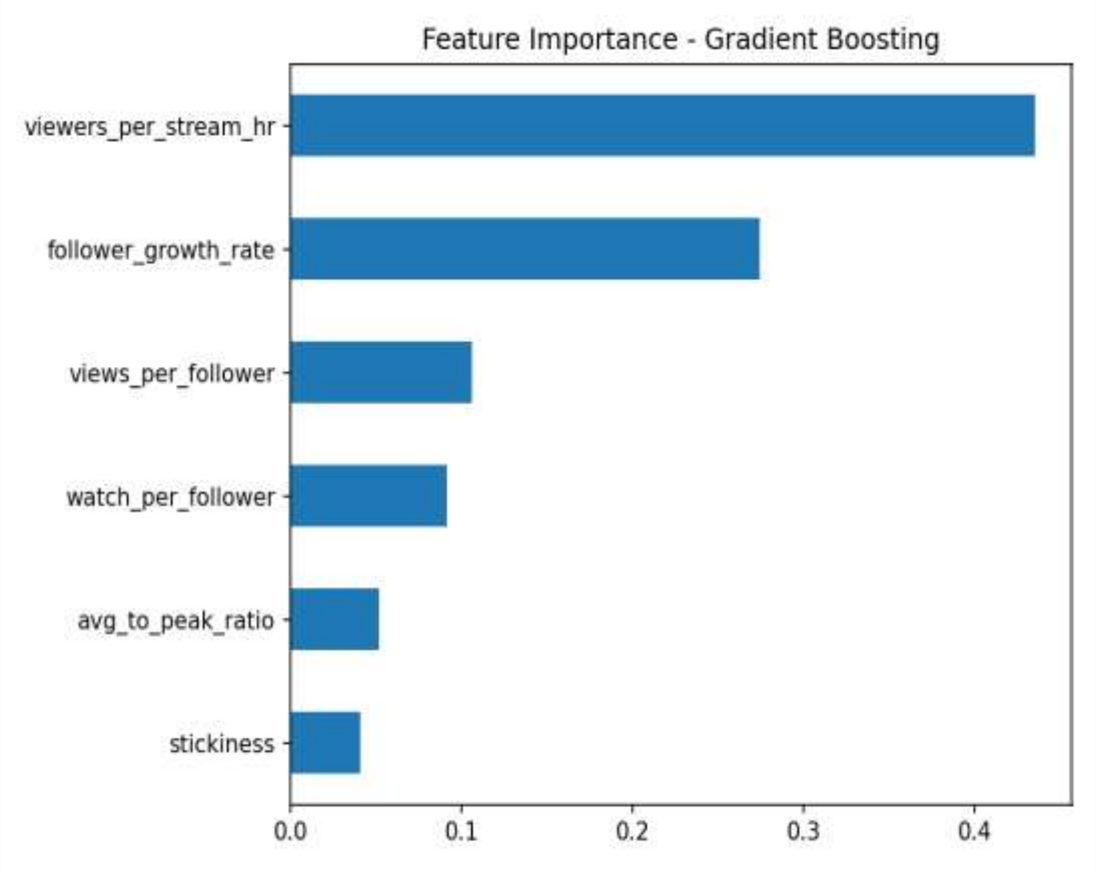
Cluster 1 — Standard

895 channels · mean EES = 0.209

The typical channel profile

Predicting Followers Gained





Gradient Boosting wins · MAE 117,977 followers

THE KEY FINDING

The efficiency–growth paradox

High-Efficiency channels **do NOT** gain the most absolute followers.

~81,000

avg. followers gained

High-Efficiency channels (Cluster 0)

~220,000

avg. followers gained

Standard channels (Cluster 1)

Nearly 3× more absolute growth for the "less efficient" tier — raw follower growth is still partly a function of channel size, not engagement quality alone.

Limitations & future work

LIMITATIONS

- Cross-sectional snapshot, not longitudinal — limits causal claims
- Equal-weighted EES — alternative weightings (e.g. PCA) untested
- No external variables — genre, sponsorships, esports events not captured

FUTURE WORK

- Time-series data to test causality (e.g. lagged regression)
- PCA-derived or expert-elicited feature weighting
- SHAP values for instance-level growth explanations

Thank you

This Concludes the Presentation.